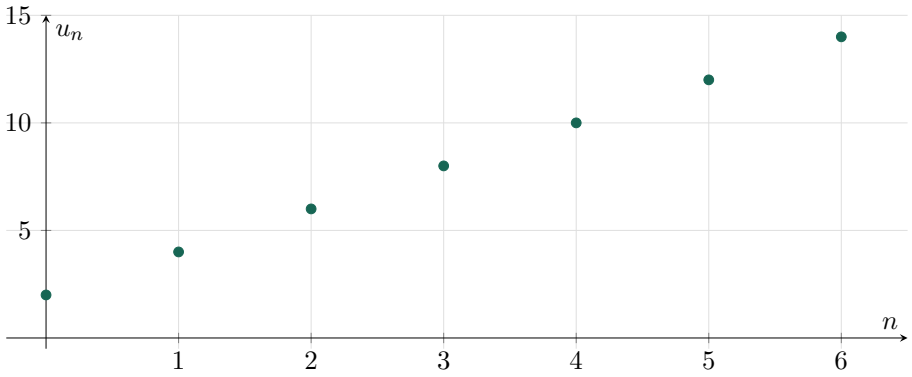
1.1 Defining and representing a sequence

Definition and results

A real sequence is a function whose variable is an integer: to each index n in the domain there corresponds a real number u_n. The notation (u_n) denotes the sequence, u_n a term. An explicit formula gives u_n directly as a function of n. A recursive definition gives a first term and a rule for getting from one term to the next. The representation is a set of points $(n; u_n)$: we do not necessarily join these points.

Status and justification. Definition. The uniqueness of the terms obtained from a first term and of a deterministic rule is assumed; no formal proof by induction is required.

Worked example. For $u_n = n^2 + 1$, $u_0 = 1$, $u_3 = 10$. For $v_0 = 1$ and $v_{n+1} = 2v_n + 1$, $v_1 = 3$, $v_2 = 7$, $v_3 = 15$. We cannot replace v_3 by $2 \times 3 + 1$: the rule refers to the previous term.



Terms of $u_n = 2n + 2$: only integer indices are plotted.

Watch out

The index is not the value. A finite list of terms does not determine a unique rule.

Immediate application. Calculate the first four terms of $w_0 = 2$, $w_{n+1} = w_n + 2n + 1$; propose an explicit formula.

1.2 Arithmetic Sequences

Definition and results

A sequence is arithmetic with common difference r if $u_{n+1} = u_n + r$ for every allowed index. Then $u_n = u_p + (n-p)r$ for $n \geq p$. If $r > 0$, it is strictly increasing; if $r < 0$, strictly decreasing; if $r = 0$, constant. The points lie on a line.

Status and justification. For p fixed, the sequence $v_n = u_p + (n-p)r$ satisfies $v_p = u_p$ and $v_{n+1} = v_n + r$. By uniqueness of the successive terms, $u_n = v_n$. The variation follows from the sign of $u_{n+1} - u_n = r$.

Worked example. A reservoir contains 140 L and receives 12 L per minute. After n minutes, $u_n = 140+12n$. At 5 min: 200 L. If we start indexing at 1 with 140 L, we write $u_n = 140+12(n-1)$.

Watch out

Adding a constant percentage does not in general give an arithmetic sequence.

Immediate application. We know $u_3 = 17$ and $u_8 = 37$ for an arithmetic sequence. Determine r and u_0.

1.3 Geometric sequences

Definition and results

A sequence is geometric with common ratio q if $u_{n+1} = qu_n$. For $n \geq p$, $u_n = u_p q^{n-p}$, with the convention that an empty product equals 1. A constant rate t gives $q = 1 + t$. The variation is determined by $u_{n+1} - u_n = (q-1)u_n$. For $q < 0$, the signs may alternate.

Status and justification. The formula gives the first term, and multiplying by q gives the next term. The uniqueness of the generation justifies the formula. For $q = 0$, all terms after the first are zero; deal with this case directly.

Worked example. A quantity of 160 mg loses 25% at each step: $u_n = 160(\frac{3}{4})^n$. The terms are positive and $u_{n+1} - u_n = -\frac{u_n}{4} < 0$. We get $u_2 = 90$ mg, without confusing a 25% decrease with a 25 mg withdrawal.

Watch out

$q > 1$ is not enough to conclude that every geometric sequence is increasing: the sign of the first term matters.

Immediate application. Study the variation of $u_n = -2 \times 3^n$, then of $v_n = 2(-1)^n$.

1.4 Sums, indices and proofs

Definition and results

For $n \geq 1$, $1+2+\cdots+n = \frac{n(n+1)}{2}$. For $q \neq 1$ and $n \geq 0$, $1+q+\cdots+q^n = \frac{1-q^{n+1}}{1-q}$. For $q = 1$, the sum is $n+1$. Thus, from u_p to u_n, an arithmetic sum equals $\frac{(n-p+1)(u_p+u_n)}{2}$; a geometric sum is $\frac{u_p(1-q^{n-p+1})}{1-q}$ if $q \neq 1$.

Status and justification. Writing $S = 1 + \cdots + n$ in both directions, each pair of the sum 2 S is $n+1$, with n pairs. For $T = 1 + q + \cdots + q^n$, subtract $qT = q + \cdots + q^{n+1}$: $(1-q)T = 1 - q^{n+1}$. The case $q = 1$ must come before the division.

Worked example. For $u_n = 3 + 2n$, $u_2 + \cdots + u_6 = \frac{5(7+15)}{2}$ = 55. For $v_n = 2 \times 3^n$, $v_0 + \cdots + v_3 = \frac{2(1-3^4)}{1-3} = 80$.

Watch out

Count the terms before using the formula: from 0 to n, there are n+1.

Immediate application. Calculate $\sum_{k=2}^{5} 2^k$ without a summation symbol in the answer.

1.5 Conjecture a limit and program a threshold

Definition and results

A limit is approached intuitively here. The terms $\frac{1}{n+1}$ approach 0; n^2 grows without bound; $(-1)^n$ does not settle down. A numerical experiment suggests a conclusion but does not prove a behaviour for all indices. A threshold algorithm must specify the direction of the inequality and plan a stopping condition.

Status and justification. Illustrative observations, without a formal theory of limits. The correctness of an integer threshold is checked by testing the index found and the previous index, when monotonicity is established.

Worked example. For $u_n = 80(\frac{1}{2})^n$, the first index such that $u_n < 6$ is 4: $u_3 = 10 \geq 6$ and $u_4 = 5 < 6$. The program begins with $n = 0, u = 80$ and repeats the division by 2 while $u \geq 6$.

Watch out

An unbounded loop may never stop if the threshold cannot be reached; $<$ and $\leq$ are not interchangeable.

Immediate application. For $u_n = 48(\frac{1}{2})^n$, determine the first indices where $u_n \leq 6$ and where $u_n < 6$.